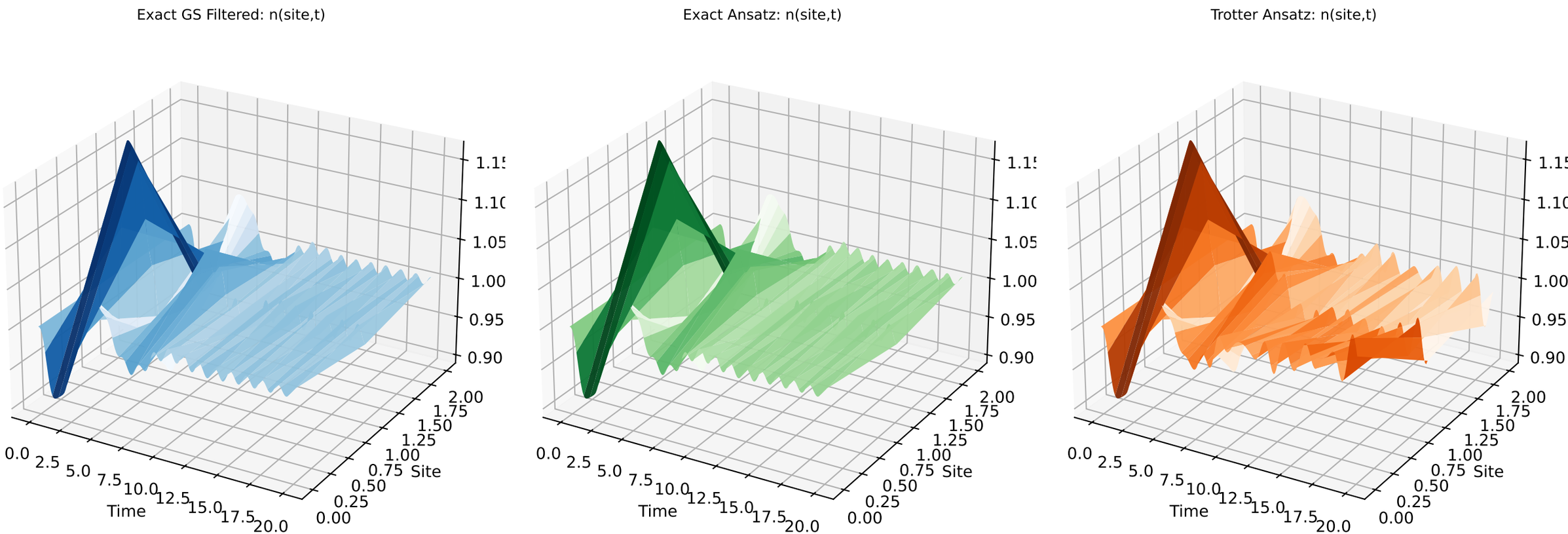
Executed Command

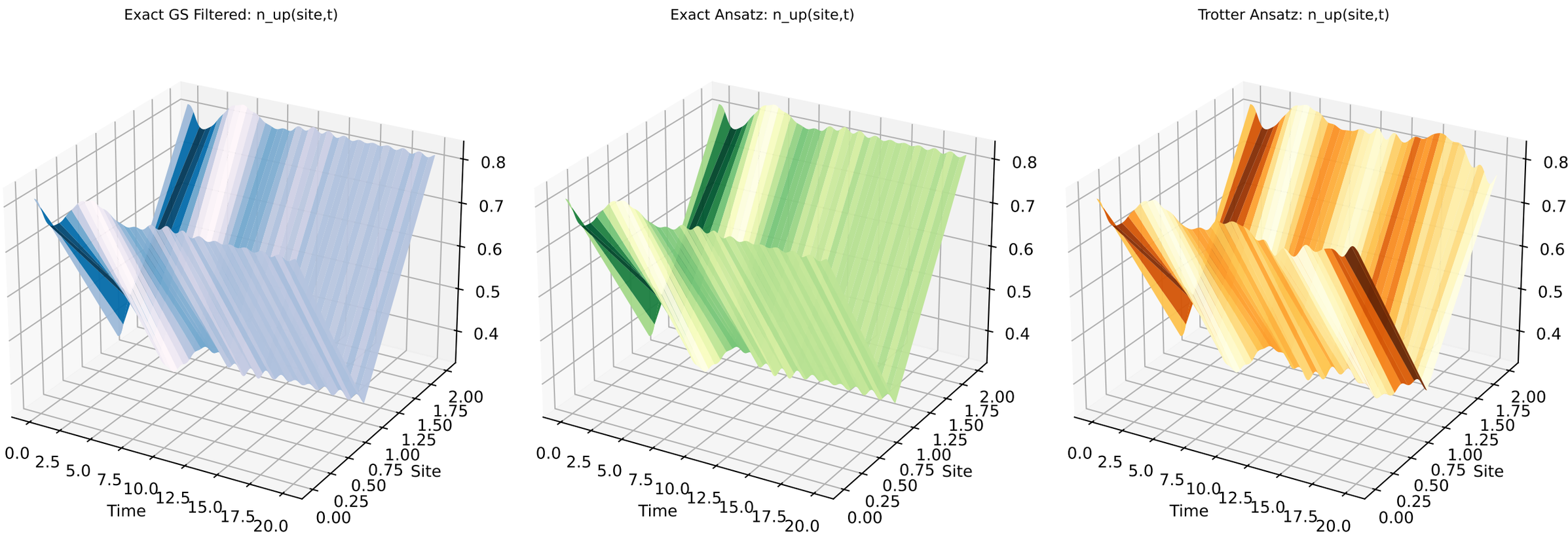
Reference: pipelines/PIPELINE_RUN_GUIDE.md
Script: pipelines/hardcoded_hubbard_pipeline.py

```
/opt/anaconda3/bin/python pipelines/hardcoded_hubbard_pipeline.py --L 3 --t 1.0 --u 4.0 --dv 0.0 --boundary open
  --ordering blocked --t-final 20.0 --num-times 201 --suzuki-order 2 --trotter-steps 64 --initial-state-source
  vqe --enable-drive --drive-A 1.0 --drive-omega 1.0 --drive-tbar 5.0 --drive-phi 0.0 --drive-t0 0.0 --drive-
  pattern staggered --drive-time-sampling midpoint --vqe-reps 2 --vqe-restarts 3 --vqe-seed 7 --vqe-maxiter 600
  --vqe-method COBYLA --skip-qpe --output-json artifacts/json/H_L3_vt_t1.0_U4.0_S64_3d.json --output-pdf
  artifacts/pdf/H_L3_vt_t1.0_U4.0_S64_3d.pdf
```

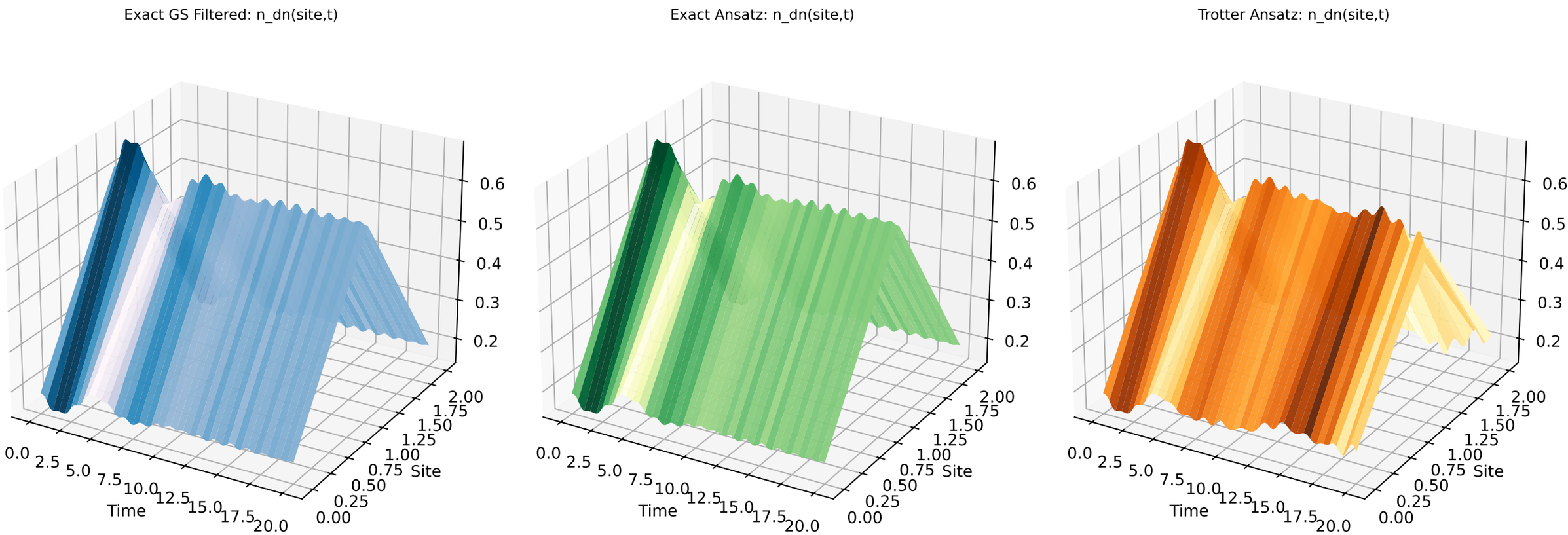
L=3 3D Densities (Total n)



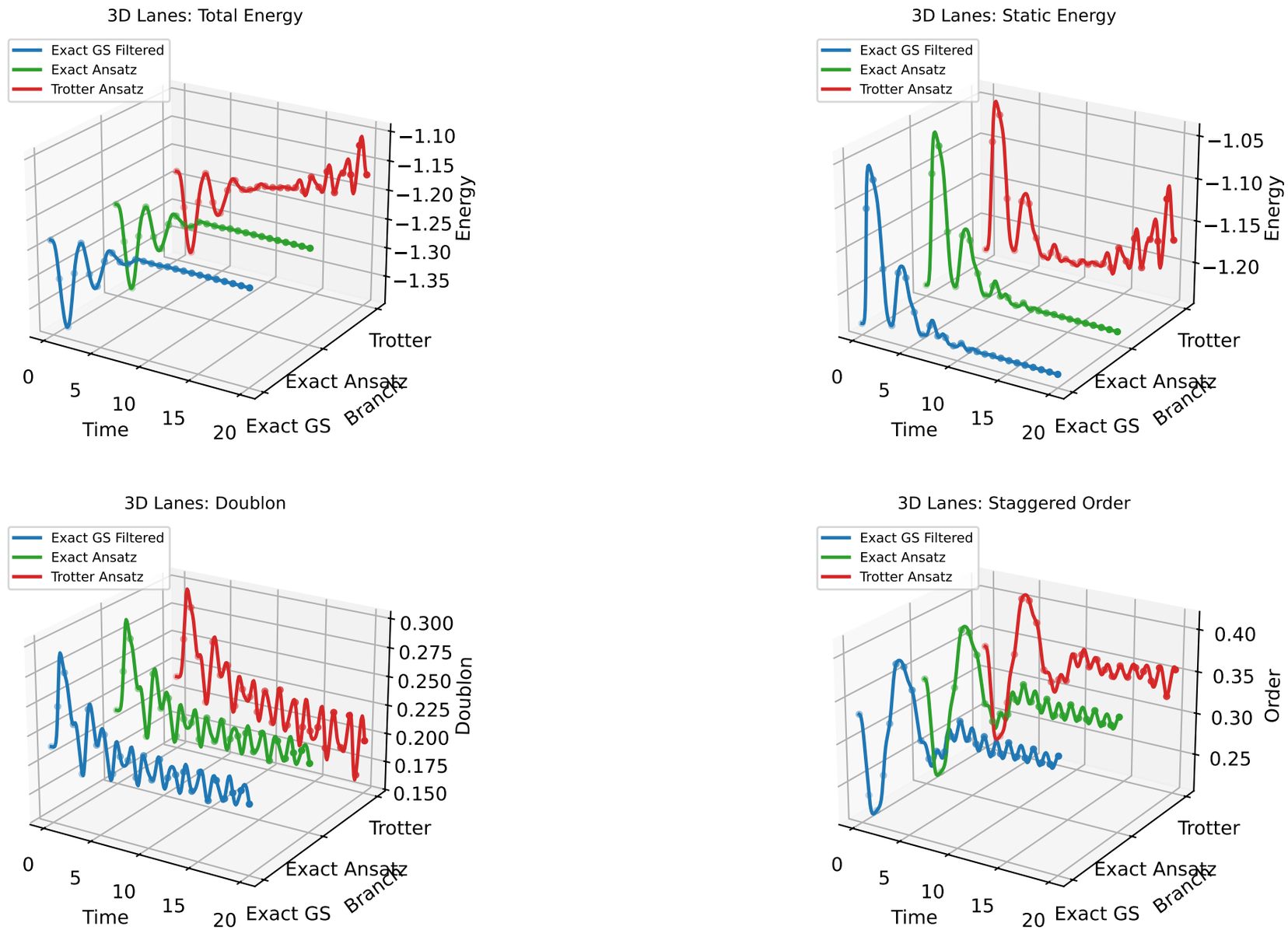
L=3 3D Densities (Spin-Up)

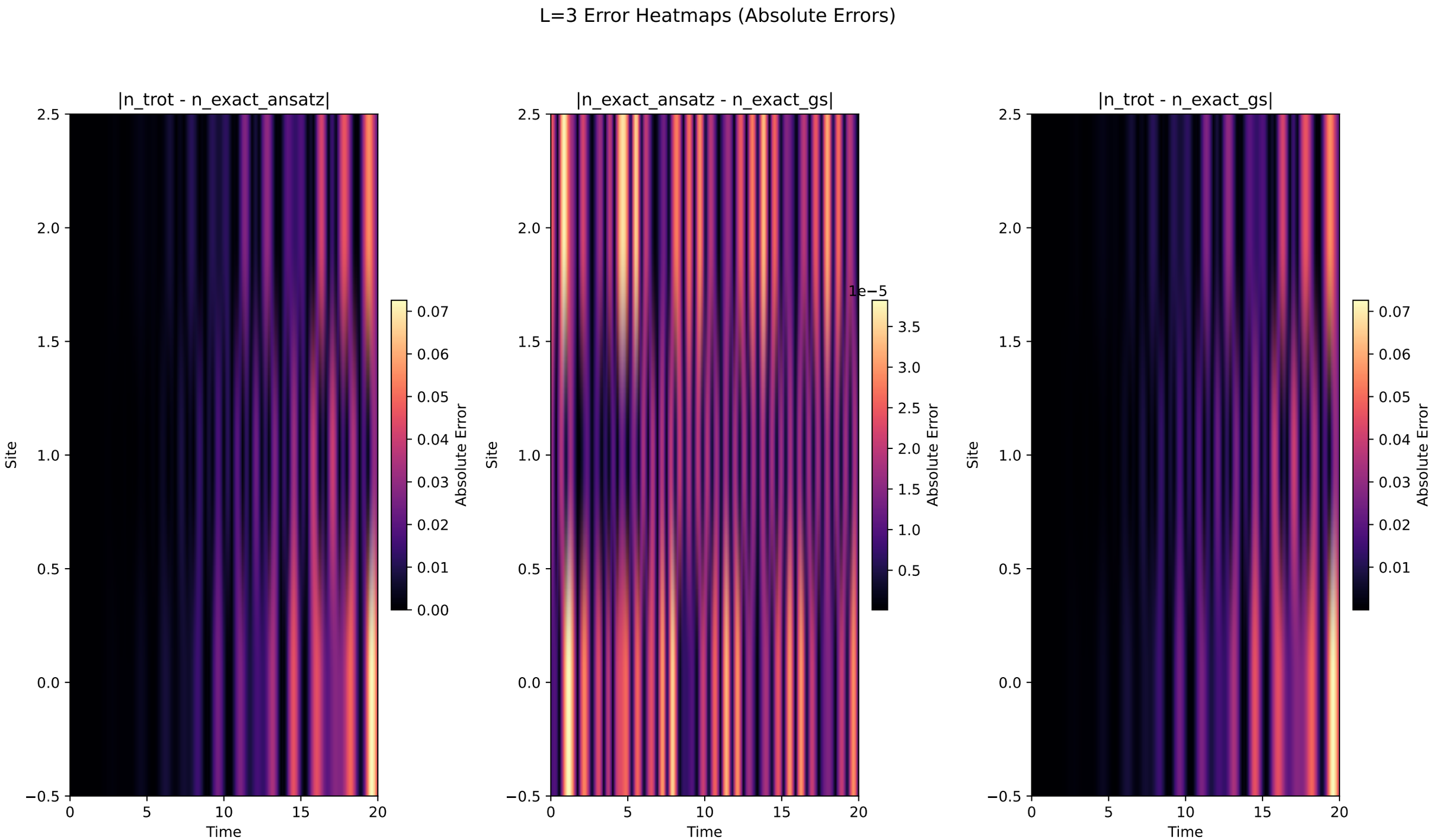


L=3 3D Densities (Spin-Down)

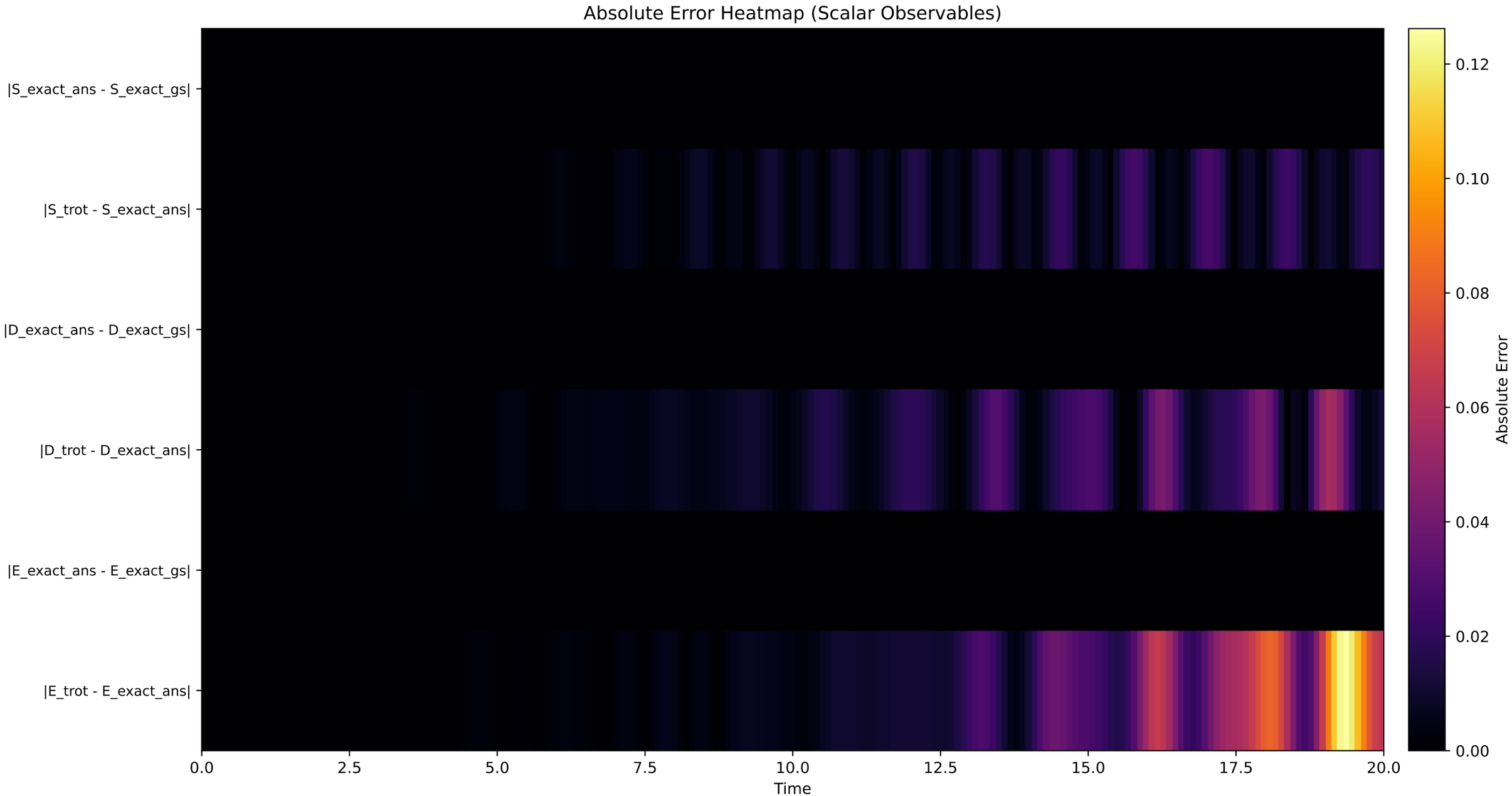


L=3 3D Scalar Observables (Three Evolutions)

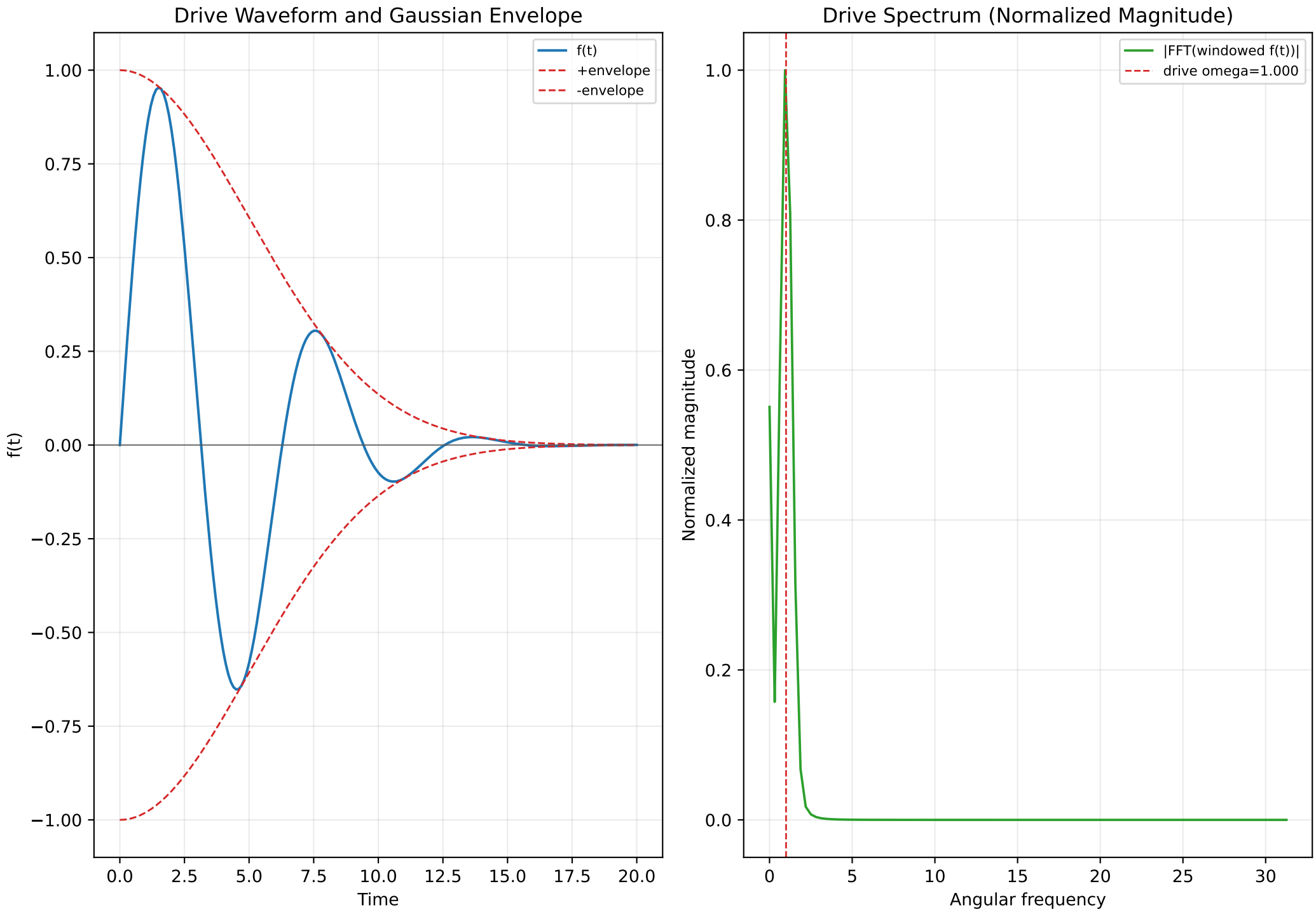


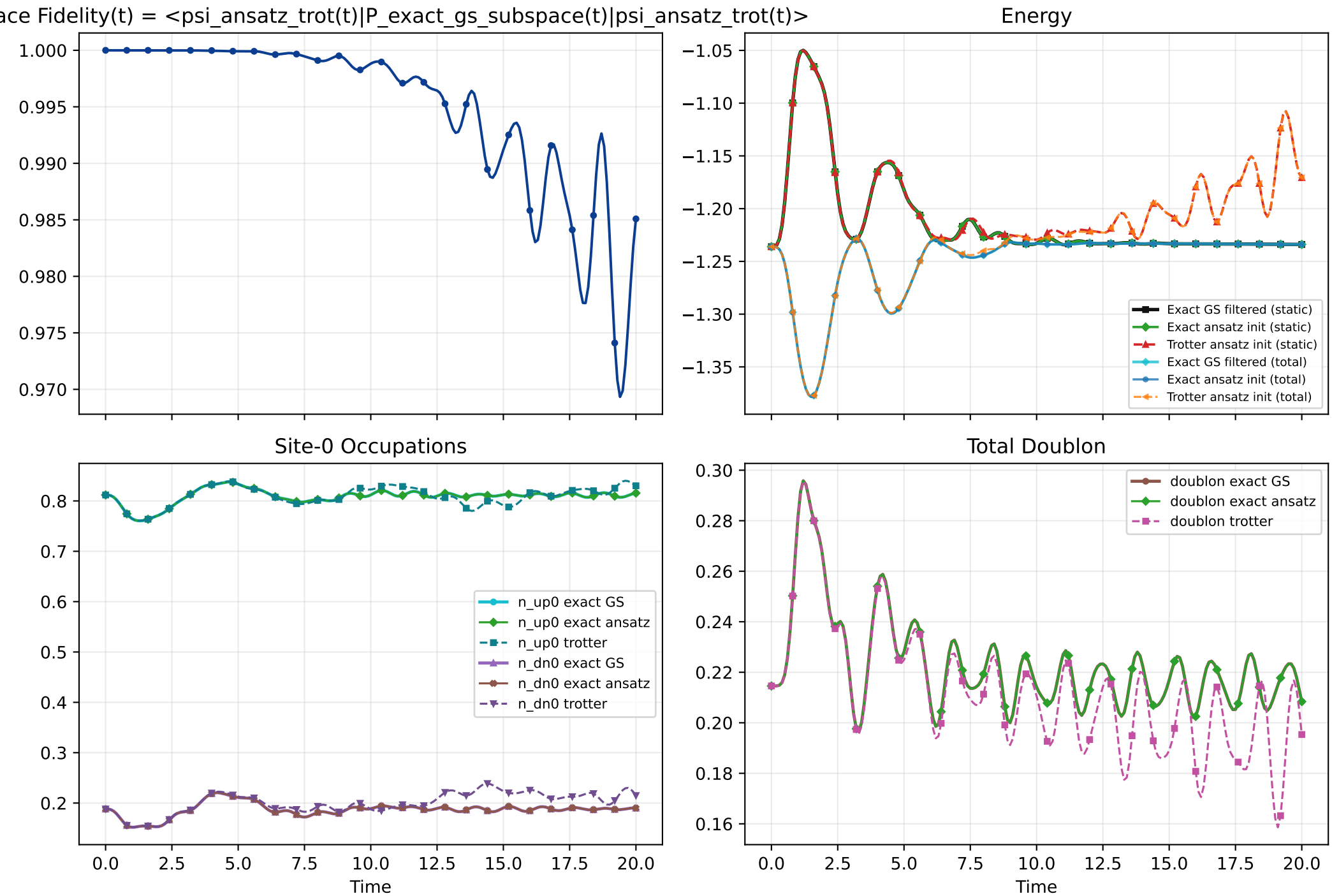

L=3 Error Heatmaps (Absolute Errors)
|n_trot - n_exact_ansatz|
|n_exact_ansatz - n_exact_gs|
|n_trot - n_exact_gs|
Site
Time
Absolute Error
1e−5

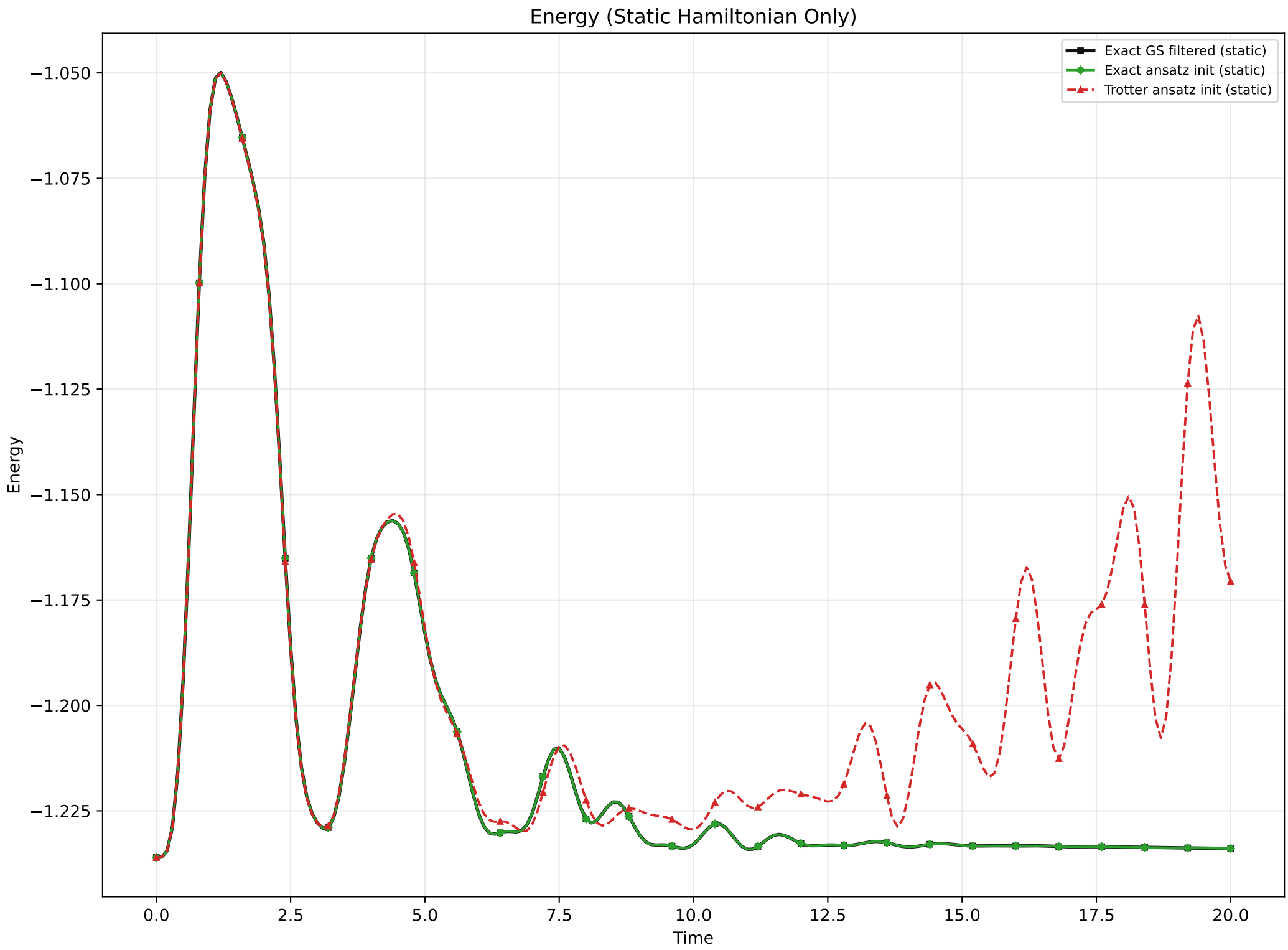
L=3 Scalar Error Heatmap

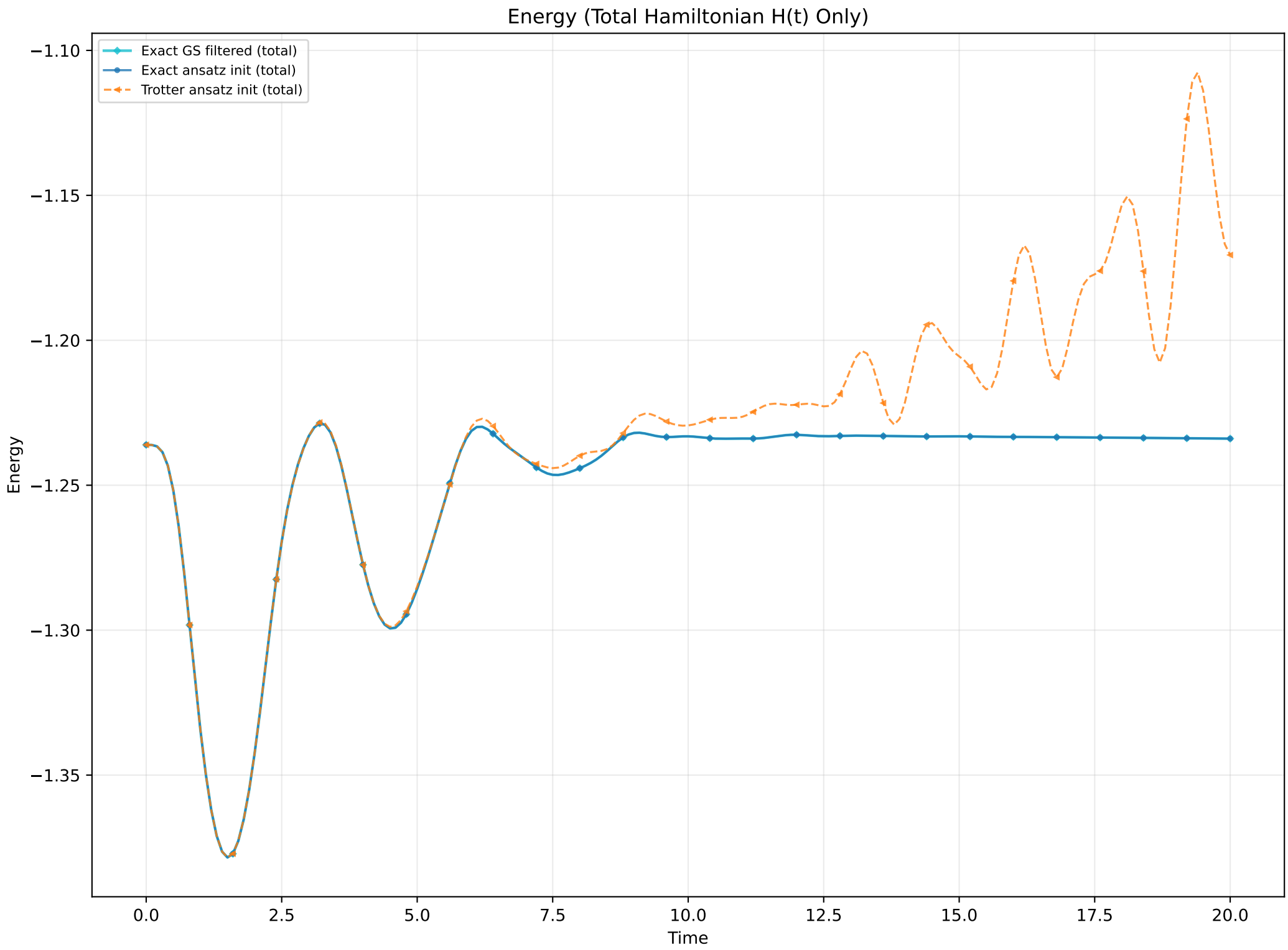


Drive Diagnostics: A=1.0, omega=1.0, tbar=5.0, phi=0.0, t0=0.0




Hardcoded Hubbard Pipeline: L=3
ace Fidelity(t) = <psi_ansatz_trot(t)|P_exact_gs_subspace(t)|psi_ansatz_trot(t)>
1.000
0.995
0.990
0.985
0.980
0.975
0.970
Energy
−1.05
−1.10
−1.15
−1.20
−1.25
−1.30
−1.35
Exact GS filtered (static)
Exact ansatz init (static)
Trotter ansatz init (static)
Exact GS filtered (total)
Exact ansatz init (total)
Trotter ansatz init (total)
Site-0 Occupations
0.8
0.7
0.6
0.5
0.4
0.3
0.2
n_up0 exact GS
n_up0 exact ansatz
n_up0 trotter
n_dn0 exact GS
n_dn0 exact ansatz
n_dn0 trotter
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time
Total Doublon
0.30
0.28
0.26
0.24
0.22
0.20
0.18
0.16
doublon exact GS
doublon exact ansatz
doublon trotter
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time

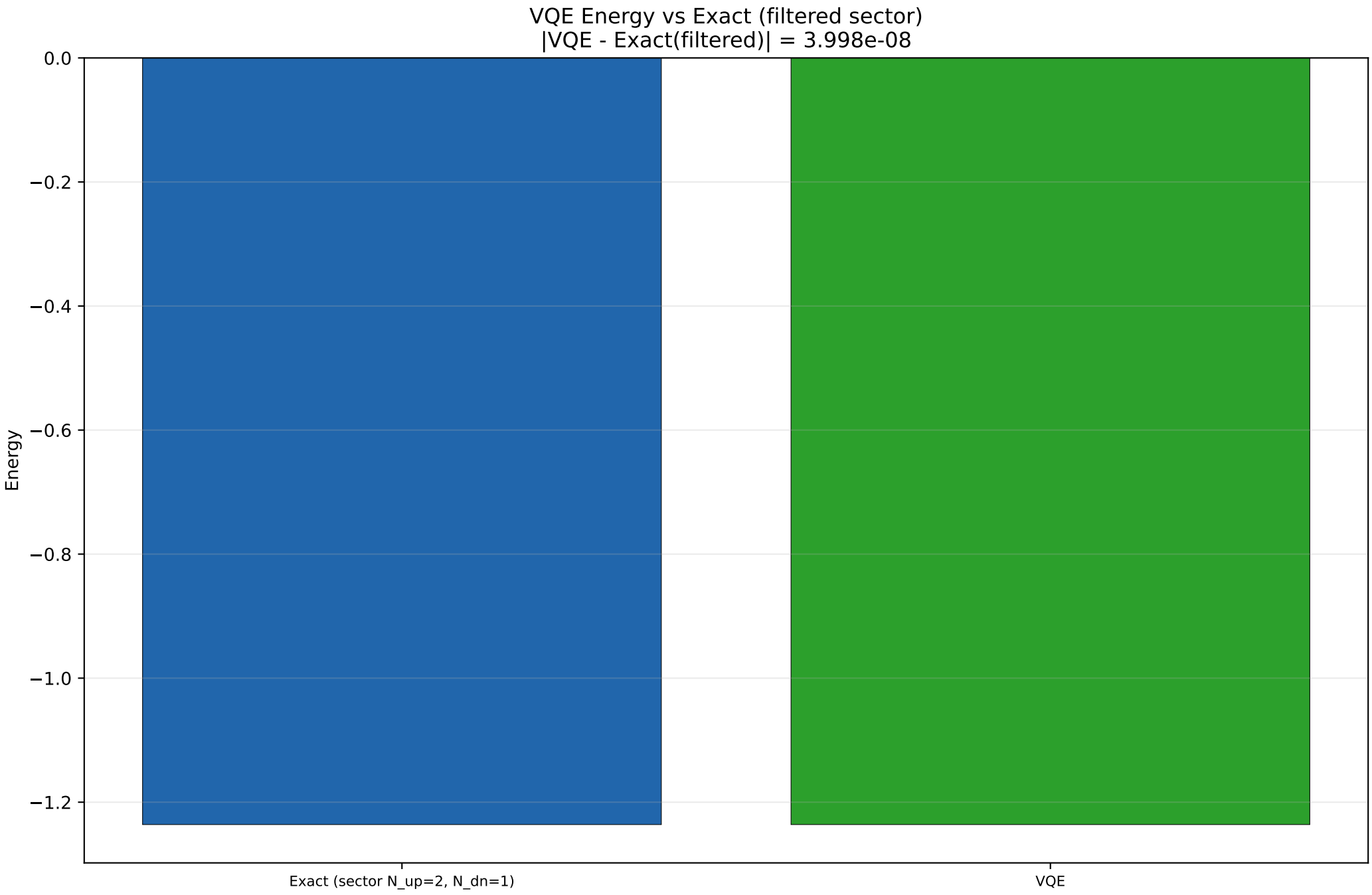

Energy (Static Hamiltonian Only)
Exact GS filtered (static)
Exact ansatz init (static)
Trotter ansatz init (static)
Energy
Time
−1.050
−1.075
−1.100
−1.125
−1.150
−1.175
−1.200
−1.225
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Energy (Total Hamiltonian H(t) Only)
Exact GS filtered (total)
Exact ansatz init (total)
Trotter ansatz init (total)
−1.10
−1.15
−1.20
−1.25
−1.30
−1.35
Energy
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time

Appendix — L=3

This section contains supporting/redundant views and metadata.
Core non-redundant analysis appears earlier in the PDF.

VQE optimises within the half-filled sector; exact (filtered) is the true sector ground state.
Full-Hilbert exact energy is in the JSON text summary only.



```
Hardcoded Hubbard pipeline summary

settings: {"L": 3, "t": 1.0, "u": 4.0, "dv": 0.0, "boundary": "open", "ordering": "blocked", "t_final": 20.0, "num_times": 201
exact_trajectory_label: Exact_Hardcode
exact_trajectory_method: python_matrix_eigendecomposition
fidelity_definition: fidelity(t) = <psi_ansatz_trot(t)|P_exact_gs_subspace(t)|psi_ansatz_trot(t)>, where P_exact_gs_subspace(t
subspace_fidelity_at_t0: 0.9999999739510665
energy_t0_exact_gs: -1.2360679774997898
energy_t0_exact_ansatz: -1.236067937517462
energy_t0_trotter: -1.236067937517462
ground_state_exact_energy (full Hilbert): -2.000000000000
ground_state_exact_energy_filtered: -1.2360679774997911
filtered_sector: {'n_up': 2, 'n_dn': 1}
vqe_energy: -1.236067937517458
qpe_energy_estimate: None
initial_state_source: vqe
hamiltonian_terms: 18
reference_sanity: {'checked': False, 'reason': 'no matching bundled reference for these settings'}
```